

WASP-50b

R-1726

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_131031 · created 2026-08-02T17:06:39+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456596.7607 +/- 0.005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.115 +/- 0.014
Transit depth (Rp/Rs)^2	1.32 +/- 0.32 [%]
Orbital Inclination (inc)	84.62 +/- 1.66
Airmass coefficient 1 (a1)	1.1481 +/- 0.0075
Airmass coefficient 2 (a2)	-0.1003 +/- 0.0078
Scatter in the residuals of the lightcurve fit is	0.66 %
Best Comparison Star	None
Optimal Aperture	3.25
Optimal Annulus	8.49
Transit Duration (day)	0.068 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.115 ± 0.014 vs 0.1404 (deviation 1.81σ)
Duration fit vs published	0.068 d vs 0.0754 d (deviation 0.41σ)
Mid-time O–C	-9.1 min
Depth SNR	8.2
Residual scatter	0.66 %

Light curve

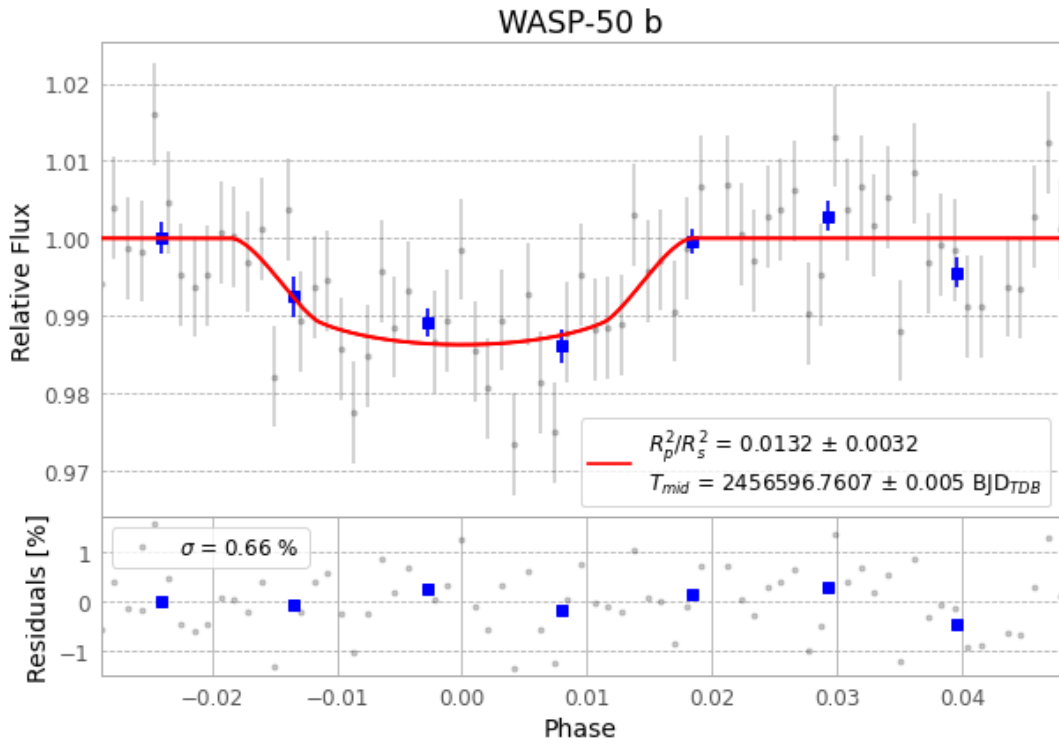


Figure 1 — FinalLightCurve_WASP-50 b_2013-10-30.png (SHA-256 615b2c0e7394b093...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [190, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5034e722cfe778ba...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), WASP-50131031044512.FITS → WASP-50131031082112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-131031.log (05e5501295b2...), FinalLightCurve_WASP-50 b_2013-10-30.png (615b2c0e7394...), FinalLightCurve_WASP-50 b_2013-10-30.csv (d7b6f630c3c9...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 17:06Z.